

Prospective weighted-CM relation search for NIST P-192 — v2 amendment

2026-10-07

Status and decision

This additive bundle approves protocol v2 for `EXP-SCURVE-1a8daf` under decision `DEC-20261007-0ed467` and the path-bound amendment; the committed v1 specification and review bundle remain unchanged historical records. **No scientific search, timing run, map census, ECDLP recovery, or replication has been performed.** Every planned run is pending and the current claim ceiling is `PROTOCOL_DESIGNED_UNEXECUTED`.

The native prerequisite is [crypto PR #1507](#), reviewed here at commit `315c7b417305db4f34ad01cb07c18ca2c55181ee`. The prior archive is [crypto-autoresearcher PR #1942](#), reviewed at `5ed7dfbbc02a975e5919bc03b3020f1931e30813`. The prior `EXP-SCURVE-647ade` is a different, immutable $L \leq 2^{48}$ generator-word census. This protocol does not extend it, rerun it, or consume its run budget.

`TASK-20261007-66d44b` is the sole archive owner for this additive payload. `CORR-20261007-f4d600` records that the v1 precursor work and review began before that ownership was bound. The remedy preserves every v1 byte, archives v2 additively, and uses a second, receipt-only commit to bind the exact first archive commit. This governance correction is not scientific evidence.

Labels used below are strict:

- **Verified (protocol):** local YAML parses, links resolve, arithmetic equalities and derived box counts pass the non-scientific checker.
- **Derived:** follows exactly from the frozen integers and equations.
- **Proposed:** an algorithm, budget, threshold, or future measurement.
- **Measured:** none.

Why this is not another blind BFS

A bounded breadth-first isogeny walk allocates effort by graph distance. It can inventory nearby curves, but it does not preferentially search for a cheap loop returning to the source. This protocol instead enumerates principal quadratic integers, whose ideal factorizations are loops by construction. It then uses a larger algebraic relation base for yield, cancels every prime above the smaller mappable base exactly, and optimizes only certified relation combinations.

That design separates four questions that are easy to conflate:

1. Does an abstract principal ideal relation exist?
2. Can its large-prime and high-degree support be cancelled exactly?
3. Can every remaining directed isogeny and the terminal isomorphism be built?
4. Does the resulting action make the same subgroup DLP cheaper after all search, construction, transfer, and verification costs are charged?

Only the fourth question can support a weakness claim.

Exact object and CM assumptions

The curve is the exact short-Weierstrass NIST P-192 / secp192r1 tuple over

P-192 weighted-CM protocol — all scientific stages pending

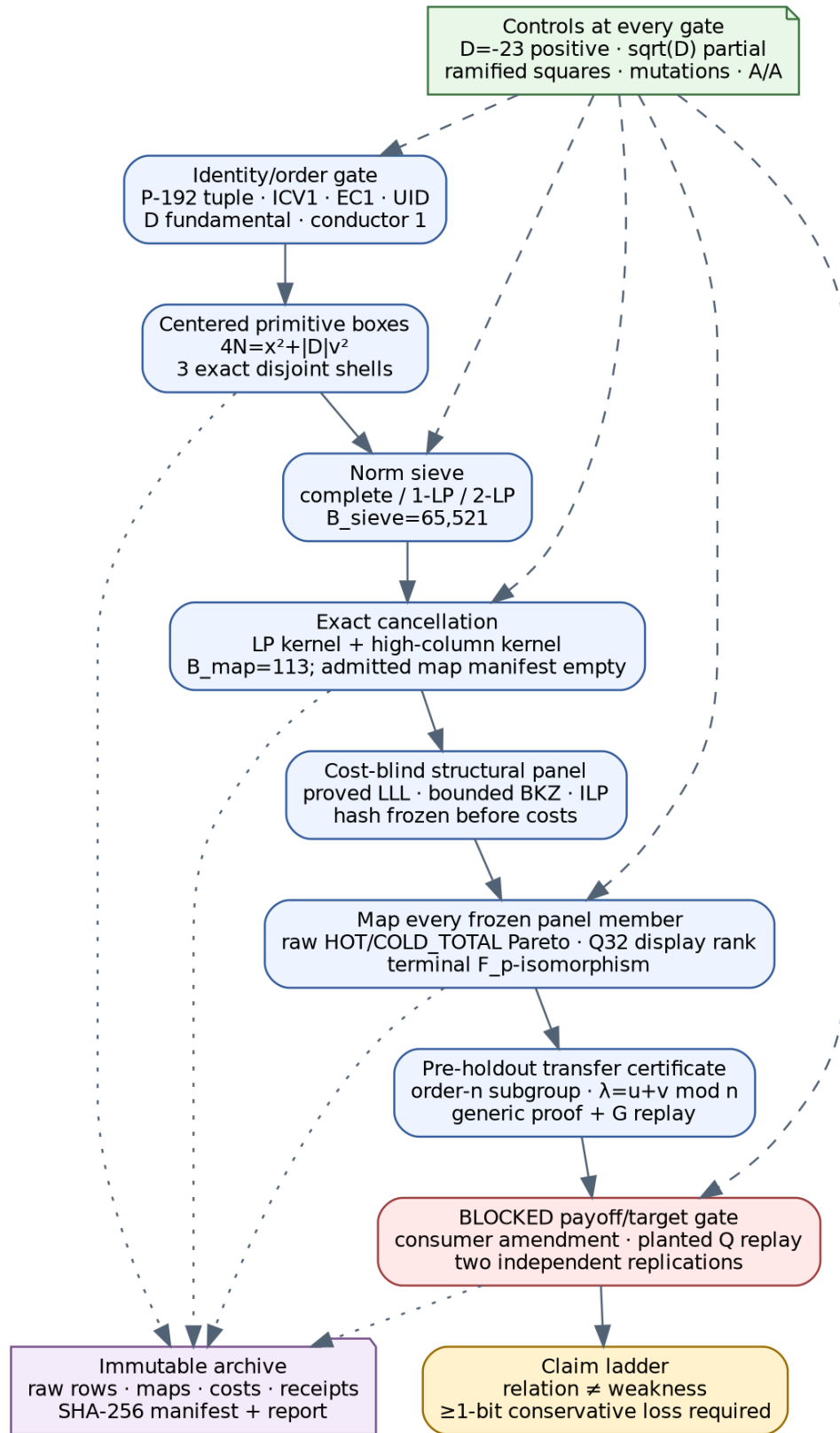


Figure 1: Frozen protocol and evidence gates

$$p = 6277101735386680763835789423207666416083908700390324961279.$$

Its coefficient is (a=p-3), its published (b,G,n,h=1) are recorded in the experiment specification, and its trace is

$$t = 31607402316713927207482677199, \quad n = p + 1 - t.$$

The three exact identity strings are:

identity	value
ICV1	icv1-fp192-t31607402316713927207482677199-52e4af59
EC1	EC1P192Cp192h5531c4a08bdb
UID	urn:ec-record:1:sha256:5531c4a08bdb64b6e86a6e30e9a08aa57e

The proposed order gate must independently replay

$$D = t^2 - 4p = -24109379060336110122544161233113975664949272517896865359515$$

and

$$D = -5 \cdot 11 \cdot 31 \cdot 14140398275856956083603613626459809774163796198179979683.$$

Write $D_{\text{pi}}=t^2-4p$ for the Frobenius-order discriminant, D_K for the fundamental field discriminant, and D_{End} for the geometric endomorphism-order discriminant. The archived derivation says the native certificate proves $D_{\text{pi}}=D_K=D_{\text{End}}=D$: p does not divide t , so the curve is ordinary, and D_{pi} is squarefree and fundamental, so $\mathbb{Z}[\text{pi}]$ is already maximal and no larger geometric order can contain it. Thus the claimed result is $\text{End}_{\text{over } \mathbb{F}_{\bar{p}}}(\mathcal{E})=\mathbb{Z}[\text{pi}]=\mathcal{O}_D$. That chain is an execution prerequisite here, not a conclusion imported from the phrase “CM discriminant” or merely from an archive URL. Since $D \leq -4$, the unit group must also be checked to be exactly $\{+1, -1\}$.

Centered deterministic search

For $\alpha = u + v\pi$, with $\pi^2 - t\pi + p = 0$,

$$N(\alpha) = u^2 + tuv + pv^2.$$

Set $(x=2u+tv)$. Then

$$4N(\alpha) = x^2 + |D|v^2, \quad x \equiv tv \pmod{2}.$$

For nonscalars, choose $v > 0, x \geq 0$. Because the only units are ± 1 , this quotients global sign and conjugation; $x = 0$ is retained as a boundary representative. Primitivity is **exactly** $\gcd(|u|, v) = 1$. Using $\gcd(x, v)$ would be wrong: the important $\sqrt{D} = 2\pi - t$ control has $x = 0, v = 2$ but is primitive because t is odd.

The inclusive boxes and exact raw counts are:

box	$1 \leq v \leq V$	$0 \leq x \leq X$	raw pairs	disjoint shell pairs
BOX-0	8	65,536	262,148	262,148
BOX-1	64	1,048,576	33,554,464	33,292,316
BOX-2	256	4,194,304	536,871,040	503,316,576

Traversal is shell, then increasing (v) , then increasing parity-compatible (x) . Logical shards contain 1,048,576 records, except a final short shard. The specification fixes domain separators, length fields, big-endian integer encoding, shard

digests, ordered shell roots, and a family root. Parallel execution cannot alter the logical bytes. This explicitly repairs the digest ambiguity identified in review of the prior archive.

For $v \neq 0$, the equation derives

$$N(\alpha) \geq \lceil |D|/4 \rceil = 6027344765084027530636040308278493916237318129474216339879.$$

The earlier (2^{48}) word boundary therefore cannot explore these principal norms. That is the mathematical reason to change representations.

Smoothness, partials, and exact recombination

The proposed algebraic sieve base contains every split or ramified prime $\ell \leq 65,521$. The final target mappable base contains the 13 analogous primes through $\ell \leq 113$, representing 23 directed orientations. The broader collection base is new; the map bound stays within the earlier degree-113 algebraic census. No source reviewed for this bundle demonstrates an admitted exact-map implementation at the former degree-4093 bound, so v2 makes no feasibility claim for that bound.

The prior P-192 kernel/codomain frontier is narrower still: artifacts cover only $\ell \in \{5, 11, 13, 23, 31, 37, 43\}$ (11 directed orientations), and those artifacts do not include the point evaluator, dual, subgroup/terminal replay, or HOT/COLD records required here. The strict admitted map-support manifest is therefore empty at protocol freeze. The remaining eligible degrees (73,89,101,103,107,113) and every source-specific direction are unavailable until exact map-and-cost admission. The bound is a maximum, not a declaration that a map exists or has zero cost.

That narrow frontier is sourced to crypto commit `aea4dbb0983a37f6463fdbde96e6884871586357`, specifically the immutable `radius-2 isogeny_routes.json` blob with SHA-256 `719846b30348a718d4369c189be74b2d7e52b8de9eb7f0be7f59176818`. It serializes kernel polynomials, Vélú codomains, and kernel-verification text through degree 43—not the stronger point evaluators, paired dual proof, subgroup transport, terminal maps, or HOT/COLD records required here. The reviewed native head `315c7b4...` contains that immutable ancestor blob.

For a split prime, the smaller root (r_0) of $r^2 - tr + p = 0 \bmod \ell$ names the positive ideal $(\ell, \pi - r_0)$; the other root is its conjugate and has negative signed coordinate. Ramified coordinates are stored modulo two, with rational scalar content recorded separately. Exact ideal valuations, including prime powers, come from ideal arithmetic/HNF—not from rational divisibility alone.

After base division, retain only:

- complete rows with residual 1;
- one-large-prime rows with a proven split prime $65,521 < q \leq 2^{31} - 1$; or
- two-large-prime rows with exactly ($q_1 q_2$), including multiplicity, both proven split primes in that interval.

The sieve never tries to factor every roughly 200-bit reject. With $L=2^{31}-1$, an exact residual $R \geq L^2$ is rejected by the one-sided inequality; only $R \leq L^2$ enters deterministic u64 factorization. Inert or missed-base factors invalidate the shell, while an unfinished exact decision is **UNRESOLVED/INCOMPLETE**. Checkpointing is allowed only at closed logical shards, with a hash-chained canonical receipt; resume must rehash every prior shard and reproduce clean-run roots and counters byte for byte.

Partials are not relations. Their signed oriented large-prime rows form an exact integer matrix (L). A graph cycle may propose combinations, but only an independently replayed arbitrary-precision solution of ($L^T c = 0$) is admitted. The same exact-kernel process cancels every algebraic-base column above 113. No high-degree edge receives an implicit zero cost.

The relation encoding is byte-exact and binds the curve UID, immutable algebraic factor-base digest, shell-local index, centered point, norm factorization, scalar content, increasing prime coordinates, explicit LP incidence, and arbitrary-precision exponents. A separate map-support manifest can evolve without changing algebraic relation IDs. One algebraic ID covers a conjugate pair, while **:positive** and **:negative** directed variants remain separate for map construction and cost.

The specification also freezes the exact RFC-8785 factor-base object keys and entry kinds, candidate/factorization byte grammars, per-candidate disposition records, family-root length fields, ordered LP/high-column coordinate manifests, and a saturated-kernel witness. The independent verifier must check $A * K = 0$, an integer left inverse $H * K = I$, and a nonzero

($m-k$)-minor; replaying the producer’s HNF routine is not independent. Checkpoint chains are per shell, bind the final protocol and producer commits, and use cumulative counters with exact conservation identities. Candidate shard bytes are regenerated from the canonical iterator on resume; disposition bytes and retained records are persisted. A partial next shard is discarded only after the last committed checkpoint validates.

Both LP cancellation and high-column elimination require saturated integer kernel bases with HNF/SNF certificates. Their kernels are infinite; only the frozen coefficient-bounded search is exhausted. For a combined vector $\mathbf{z}$, the verifier forms the exact oriented ideal $J(\mathbf{z})$, finds a normalized primitive $\mathbf{beta}=\mathbf{a}+\mathbf{b}\cdot\mathbf{pi}$, and proves $\text{HNF}(J)=\text{HNF}(\mathbf{beta})$ and $N(\mathbf{beta})=\text{product_split}(\mathbf{e11}^{\sim}\text{abs}(\mathbf{z_e11}))\cdot\text{product_ram}(\mathbf{e11}^{\sim}\mathbf{r_e11})$, where each ramified $\mathbf{r_e11}$ is the projected bit in $\{0,1\}$, not its pre-projection signed valuation. It does not assume a formal quotient of source generators is already the primitive generator.

Structural solvers, maps, and post-map weighting

Map costs do not exist when the relation lattice is first available. To avoid a circular search, the exact row lattice is first explored without HOT/COLD, Q32, map availability, degree medians, or timing inputs by:

- proved LLL with ($\delta=99/100, \eta=51/100$), 256-bit precision, and a 10,000,000-swap cap in forward and reverse row order;
- deterministic single-thread BKZ at block sizes 20, 30, and 40, two tours and 10,000,000 enumeration nodes per tour; and
- deterministic single-thread ILP with $|c_i| \leq 2^{20}$, 50,000,000 nodes, zero requested gap, and exact replay of every feasible result.

Solver exhaustion is censored. **optimal** is permitted only with an independently checked zero-gap/proof record; otherwise the label is **BEST_FOUND_WITHIN_BUDGET**. The finite retained union is exact: all rows from the two final LLL bases; all rows from every completed BKZ tour snapshot; and the final feasible incumbent, if any, from two structural ILP schedules in both row orders. Both directed variants are deduplicated and sorted by ID, then **structural-candidate-panel.json** and its checksum are closed before role 6 opens any cost artifact. No arbitrary top-k or observed map outcome filters the panel.

Role 6 calibrates Q32, then attempts **every** frozen directed panel member in (**relation_id,directed_variant_id**) order—not Q32 order. Each receives one terminal disposition. Each disposition hashes its same-index canonical evidence sidecar and, for a realized composite, carries the full ordered array of map-support record indices with repeated edges retained. Only after complete coverage does the protocol form the raw Pareto layers of realized members separately for the six-component HOT tuple, the six-component COLD_TOTAL tuple, and their joint 12-vector. The selected vectors are exactly joint layer zero. Q32 supplies separate HOT and COLD totals and only breaks display order lexicographically after the three raw layer indices. A timeout or missing disposition is **INCOMPLETE**; no selected set claims global or lattice-wide cost optimality.

Each realized directed edge and composite reports the raw tuple

$$(\text{FpM}, \text{FpS}, \text{FpI}, \text{FpAddSub}, \text{polyCoeffM}, \text{logicalBytesRW}).$$

Role-6 HOT is online evaluation and required conversions/checks on (G). The generic subgroup proof plus (G) is the pre-holdout control; there is no substitute curve-point panel and no planted target. Role 7 separately certifies every revealed planted (Q) and supplies whole-attack accounting. **COLD_SETUP** is root/kernel/map/dual construction, verification, serialization, and cleanup from an empty experiment cache. Every record must satisfy componentwise

$$\text{COLD_TOTAL} = \text{COLD_SETUP} + \text{HOT}.$$

The source UID, target UID, root, orientation, kernels, maps, and endpoints key an edge record. A median cost for a degree or an undirected prime is never a measured path cost. Relation discovery and solver cost remain separate and are included in a whole-method cold comparison unless an explicit public precomputation reuse count is also reported.

Q32 weights normalize FpM to $(2^{\sim}\{32\})$, use ties-to-even rounding, and are derived from 16 deterministic cases with one warmup and 11 ABBA repetitions. They are **ranking only**. They cannot alter membership, acceptance, stopping, or a security claim. The fixed calibration panel receives medians, MAD, IQR, range, and leave-one-case-out rank sensitivity—not an iid population interval.

The map tuple is not treated as a whole-attack ledger. A future consumer must count, with non-saturating integers, group and scalar-ring arithmetic, canonicalization/action calls, encodings, hashes, distinguished-point storage, cycles/restarts, collision recovery, and verification, as well as maps. Q32 cannot create a scalar advantage ratio. The consumer amendment must freeze measured CPU cycles as the implementation-scoped ratio plus a separate conservative full-order valuation. Componentwise dominance remains diagnostic and cannot satisfy this version's scalar-ratio claim tiers.

Map, subgroup, and terminal certificates

Every promoted candidate must archive:

1. every source/target curve identity and coefficient tuple;
2. **e11**, both roots, a global orientation proof that $\ker(\mathbf{phi})$ is the separable cyclic order-**e11** subgroup scheme $E[\mathbf{e11}] \cap \ker(\mathbf{pi}-[\mathbf{r}]) = E[(\mathbf{e11}, \mathbf{pi}-\mathbf{r})]$ on the actual source—not the generally much larger full kernel of $\mathbf{pi}-[\mathbf{r}]$ —plus kernel polynomial, rational map, exceptional divisor, and dual-composition identity;
3. every intermediate model and the final $\mathbf{F}_p$ -isomorphism back to the exact source UID—not merely an equal (j)-invariant;
4. the exact ordered edge circuit—not an infeasible expanded degree-about- 2^{192} rational function—normalized primitive $\mathbf{beta}=(\mathbf{a}, \mathbf{b})$, norm, relation, independent terminal unit signs, and both conjugate eigenvalues;
5. pre-holdout subgroup replay on (G), including dual identities and $\psi(G) = [\lambda]G$; and
6. only in role 7, a separate planted-point certificate for every revealed $Q = [d]G$, including $\phi(Q) = [d]\phi(G)$, dual identities, and $\psi(Q) = [\lambda]Q$.

The transfer cost includes forward maps, applicable duals, conversions, exceptional handling, terminal isomorphism, validation, and return to the source representation. Moving a DLP to an isogenous curve does not make it easier by itself.

No primary consumer is selected or implemented in this protocol version. An eligible future primary consumer may be an explicitly implemented automorphism-quotiented rho walk. A certified order of **lambda** requires a factorization/order certificate for **n-1**; order unknown blocks the orbit-payoff lane. The walk must prove equivariance and charge orbit canonicalization, map/inverse evaluation, partitioning, cycle detection, distinguished points, exceptions, and recovery. No ideal square-root-of-action-size gain is assumed. Explicit orbit enumeration is capped at 64; larger actions need a separately certified non-enumerative consumer. A decomposition alternative must certify a balanced basis of $\langle (\mathbf{n}, 0), (-\mathbf{lambda}, 1) \rangle$ and charge the actual multi-scalar algorithm. An eigenvalue or determinant alone is not a speedup.

The exact candidate and generic walks are deliberately an open pre-dispatch gate: a versioned amendment must freeze state/coefficient invariants, partition and update rules, increment derivation, canonicalization and baseline negation, distinguished points, collision recovery, success/termination, memory, parallelism, targets, and whole-attack accounting. Until then, the planted panel is functional/map holdout evidence only and cannot produce a DLP speedup or weakness label. In particular, a full-group rho walk remains an order-**n** computation even when the planted scalar is below 2^{36} . The frozen small- interval targets use interval BSGS only for recovery/transfer correctness; a future performance amendment must name and prove either an interval-preserving candidate consumer or a separately modeled full-group consumer.

Controls and statistical confidence

Positive controls include the (D=23) order-three norm-2 relation. The P-192 $\sqrt{D} = 2\pi - t$ row is a deliberately partial control: its factorization is $5 \cdot 11 \cdot 31 \cdot C$, algebra must pass, and map/payoff must reject its unavailable (C)-degree support and scalar action on $E(\mathbf{F}_p)[n]$. Ramified squares, inert primes, wrong orientations, unmatched large-prime columns, twist/UID substitutions, subgroup mutations, and map mutations are negative controls.

The scalar reference scan and segmented sieve must agree byte for byte on a small admitted box. Forward/reverse enumeration and one-worker/eight-worker sharding must recompose to identical roots. All-ones, shuffled, and perturbed Q32 weights may move only post-map display ranks; they cannot change accepted relation IDs, the frozen structural-panel hash, map-attempt order/dispositions, or raw Pareto dominance. A/A timing and operation controls must be consistent with parity.

The deterministic census is a finite population: report exact counts and digests, with no binomial interval and no

extrapolation outside the boxes. Smoothness-model comparisons are descriptive. The current 31-target panels at interval widths 28, 32, and 36 are functional and have no ratio estimand. Targets use a coordinator salt commitment made before selection; the salt is revealed only after one candidate, binary, and consumer are frozen. Exact domain-separated SHAKE256 bytes derive each nonzero $\mathbf{d}$, the solver receives only $(\mathbf{Q}, \mathbf{bound})$, and $\mathbf{d}$ is disclosed only after its output digest commits. Timing, ABBA order, a primary rung, bootstrap intervals, and success intervals remain disabled until the exact-consumer amendment freezes matching interval or full-group semantics. Any later claim also requires a fresh clean rebuild on two distinct isolated hosts or environments, with no shared build/cache workspace, to agree in direction.

Calibration, relation discovery, and solver outputs are selection data. Exactly one directed candidate and consumer are frozen before planted target scalars or timing-order material are derived. More than one holdout candidate requires a new preregistered familywise rule; choosing the fastest afterward is invalid.

Claim ladder and stopping rules

- **ABSTRACT_RELATION_FOUND**: exact principal row plus independent ideal/HNF replay.
- **DIRECTED_MAP_CHAIN_VERIFIED**: complete explicit maps, terminal isomorphism, and generator-subgroup action; planted-point replay remains a role-7 gate.
- **RANKED_CANDIDATE**: realized joint raw Pareto layer plus Q32 display order; no speed claim.
- **SCOPED_ADVANTAGE**: exact consumer gate closed; HOT and COLD paired 95% upper endpoints below 1; 31/31 primary functional successes; controls and both reserved independent replication instances.
- **MATERIAL_WEAKNESS_CANDIDATE**: additionally conservative HOT and COLD ratios at most 0.80 with a verified action mechanism.
- **WEAK_CURVE_WITHIN_SCOPE**: full-order derivation gives at least 1.0 conservative security bit loss, ratio at most 0.5, interval lower bound above 1 bit, and both independent replication instances.

A bounded planted solve alone never establishes full-order P-192 weakness. A timeout, OOM, missing map, solver cap, or partial shell is **INCOMPLETE**. A valid full null can say only **NO_WEAKNESS_FOUND_WITHIN_SCOPE** for these boxes, bases, bounds, solvers, and maps.

The family has eight frozen pending roles: preflight/controls, three sequential box shells, exact recombination plus a cost-blind structural panel, exhaustive frozen-panel map/cost replay plus post-map ranking, a blocked planted panel, and blocked independent rebuild/replication. Roles one through six are composite stages whose producer bytes must pass a separately built verifier before advancement. The seventh and eighth roles have no executable argv until the exact-consumer amendment binds the candidate, map manifest, salt commitment, functional interval solver, and two exact independent replication invocations. Canonical RUN IDs are allocated and rechecked only at dispatch; content-only names are not reservations. All are pending. A new Coordinator dispatch decision must bind final clean source and protocol commits before any run manifest exists.

Requirement-to-evidence matrix

- **Experiment, decision, and run family**: ledger records, experiment spec, and `run-family.yaml` are complete as protocol; every role is pending.
- **Exact P-192 tuple and identities**: `inputs.curve` is specified; execution replay is pending.
- **Discriminants, factorization, and maximal order**: `inputs.cm_order` contains the local identities; native certificate replay is pending.
- **Centered boxes and ordering**: `centered_norm_search` is specified and its raw counts pass the static checker.
- **Norm, smoothness, and LP rules**: `factor_bases_and_smoothness` is specified and unexecuted.
- **Encoding, orientation, and recombination**: `relation_encoding_and_recombination` is byte-exact; implementation is pending.
- **HOT/COLD and Q32**: `cost_contract` is specified; there are no calibration data.
- **LLL/BKZ/ILP**: `candidate_generation_and_post_map_ranking` freezes the cost-blind panel and post-map ordering; no solver or map ran.
- **Maps, subgroup, and terminal certificates**: schemas are specified; there is no candidate.
- **Controls, stop, and claim rules**: frozen but not exercised; optional norm-bit/v diagnostics are omitted unless a versioned pre-dispatch amendment freezes their boundaries, and their omission does not block the exact census.
- **Archive outputs**: the protocol bundle and SHA256SUMS are present; all scientific outputs are pending.

Canonical-knowledge disposition

The current `knowledge/frontiers/ecdlp` material and the P-192 archive PR were checked as context. No canonical graph or knowledge finding changes here: there is no new measured relation, map, speedup, or obstruction. Promotion is expressly deferred until an independently replayed run supports a scoped claim.

Immediate gate

The next action is implementation admission, not a scientific search: add the segmented sieve, exact independent ideal verifier, partial/high-column kernels, directed map and cost records (first for the currently demonstrated degrees through 43, then degree-by-degree through 113), optimizers, supervised runner, and independent replayer to the native repository; freeze the exact matched DLP consumer in a versioned amendment; pin and review the commits; then create a fresh dispatch decision. Until those gates, every result field remains empty.